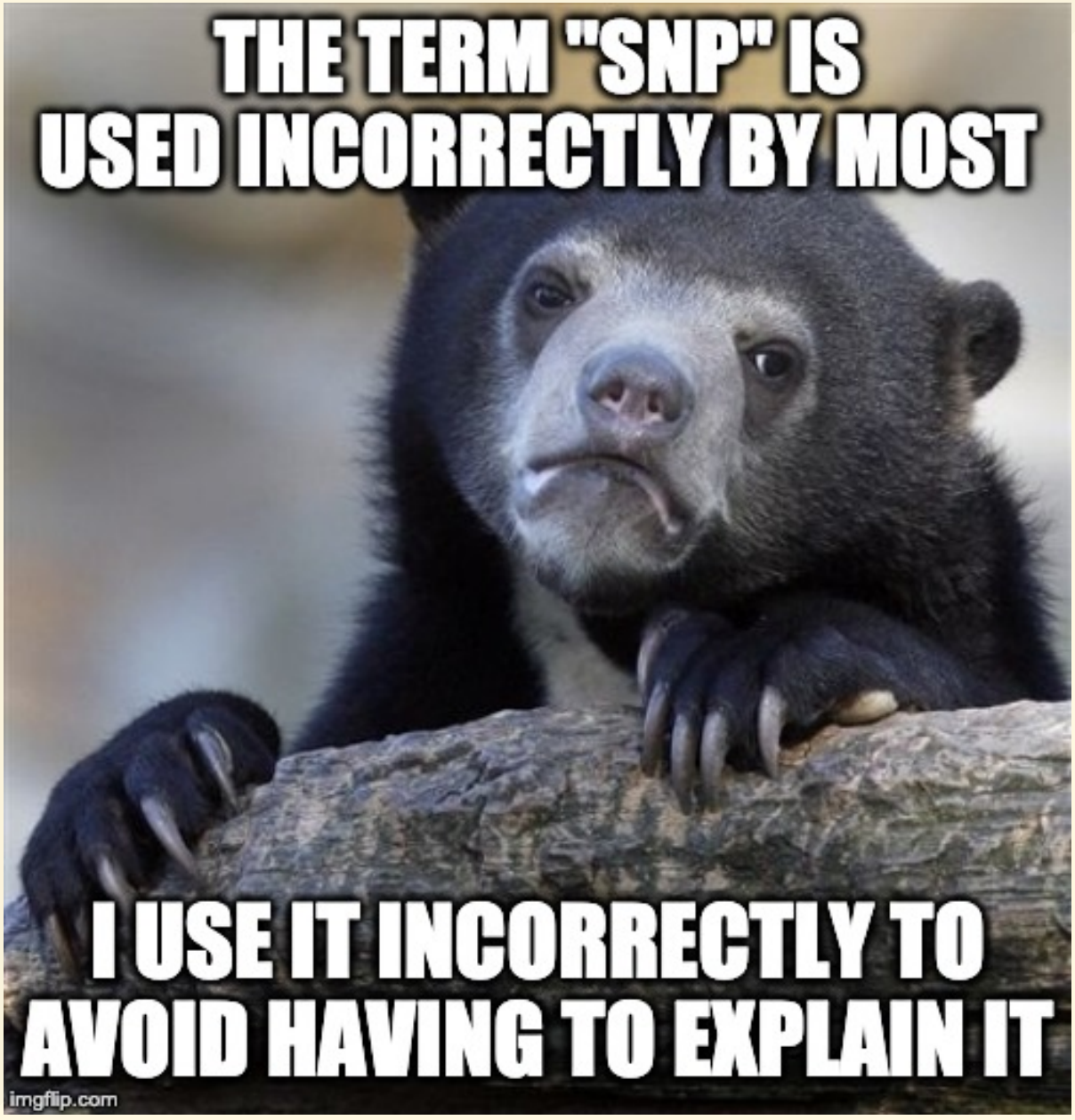
Detecting short genomic variation

a.k.a SNP calling

What is a SNP?


THE TERM "SNP" IS
USED INCORRECTLY BY MOST
I USE IT INCORRECTLY TO
AVOID HAVING TO EXPLAIN IT
imgflip.com

What is a SNP?

- **SNP**, pronounced *snip*; plural *snips*

A DNA sequence variation that **occurs commonly** within a population (e.g. 1%) where a single nucleotide: A, T, C or G of a shared sequence differs between members of a biological species or paired chromosome.

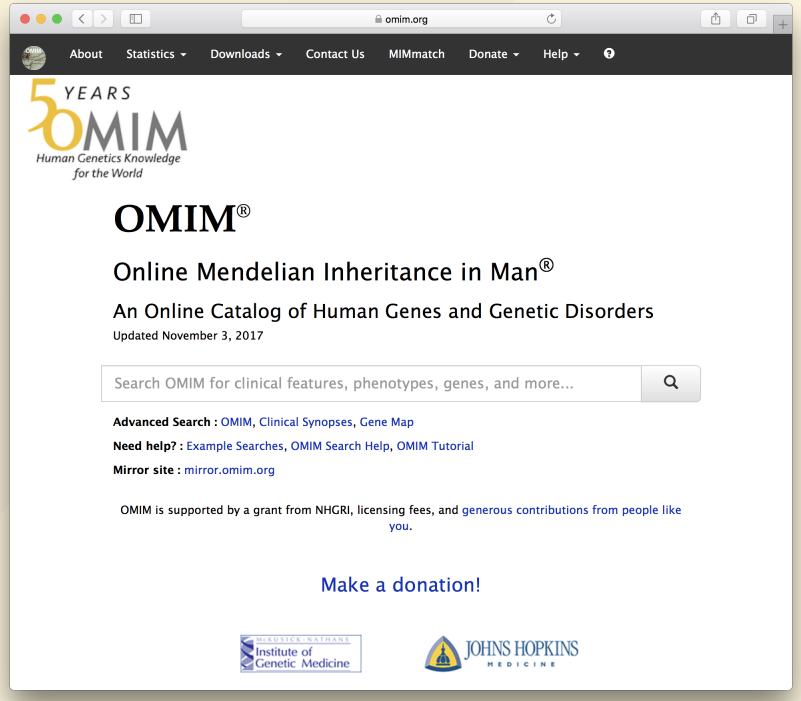
Biologists **love** SNPs because they hope for an **easy** explanation to whatever they are looking for:

Well, it's a SNP! Publish it! Case closed. Next!

What can be directly explained by SNPs?

What can be explained by variation?

How many phenotypes with known genetic explanations? See: **Online Mendelian Inheritance in Man**



See book chapter on

OMIM for more information.

So what is a variant?

- **SNPs**, single nucleotide polymorphisms. The difference consists of a single base change.
- **INDEL**s, insertions and deletions. The difference consists of having a single base added or removed.
- **SNV**s, single nucleotide variants. A combination of SNPs and INDELs. The difference is a single base.
- **Short** variations. Variations typically less than `50bp` in length.
- **Large-scale** variations. Variations larger than `50bp`
- Genomic **rearrangements**. Typically variations on the kilo-base (`1000bp`) scale.

What about in between?

Who decides what is short, long?

What if something is right at the cutoff of short and long?

Technology vs Reality

The technological limitations (read lenghts, read accuracy) determine what variations can be detected with it. Everything else is **could be lost** from sight!

The **streetlight effect** dominates. People look where there is light because things are visible there.

We are early in the process understanding what we can reliably detect.

Great news: there is **plenty left to find**! You can use your eyes and mind to find what everyone else missed!

What is ploidy?

The number of sets of chromosomes in a cell.

The number of possible alleles for a sequence.

Common terminology:

Haploid (1), diploid (2) and polyploid (3 or more).

Reality is more complicated (as always). Not all chromosomes must have the same number of copies. Example human sex chromosomes (X and Y).

More terminology: for diploid organisms

- **Homozygous**: two identical alleles at a given locus.
- **Heterozygous**: two different alleles at a locus.
- **Hemizygous**: only a single copy of a gene in an otherwise diploid organism.
- **Nullizygous**: both copies of a gene are missing in an otherwise diploid organism.

Viewing variations. The pileup.



What is a pileup?

Samtools multi-sample pileup: all bases at an

```
samtools mpileup
```

Needs a:

- A SAM/BAM alignment file.
- A genome reference file (optional).

Can produce pileups as well as genotype likelihoods.

How likely is that a given index is covered by an A,T,G,C, deletion or insertion)

What does a pileup look like?

Create a pileup:

```
samtools mpileup -f $REF demo.bam  | head
```

Prints:

```
AF086833.2 46 G 2  ^].^].       @b
AF086833.2 47 A 3  ..^],        <]B
AF086833.2 48 A 3  ..,         @cE
AF086833.2 49 T 3  ..,         FgH
AF086833.2 50 A 3  ..,          FdF
AF086833.2 51 A 5  ..,^].^]. Fg@C@
AF086833.2 52 C 7  ..,..^].^]. DgGCBCC
AF086833.2 53 T 7  ..,.... DgHC@CC
AF086833.2 54 A 7  ..,.... HiEFFCC
AF086833.2 55 T 7  ..,.... HkHFFFF
```

Weird symbols that explain what happens at each coordinate

Most SNP calling algorithms operate on pileup information.

Why all those details?

Understanding alignments is key to understanding variants.

You want to make sure the alignment is correct - or that the call is right.

- `.` a match on the forward strand
- `,` a match on the reverse strand
- lower case letter, a mismatch on the forward strand
- capital letter, a mismatch of the reverse strand

Pileup examples. What does this mean?

```
G..,....gg,,...GG,,.gg
```

- 10 bases match on the forward strand,
- 5 bases match on the reverse strand
- 4 bases indicate a mismatch of `g`
- 3 bases indicate a mismatch of `G`

15 bases indicate no mismatch, 7 bases indicate a mismatch and all agree on the mismtch (G).

Would you trust this variant? Is this homozygous, heterozygous? Something else?

Starts/ ends of alignments are more suspicious

Sequences misalign more frequently at starts and ends.

Special characters indicate bases that align at start and end.

- `^` and `$`

See the book course page for links to full description.

A variant caller reads off all these signals and tries to reconcile them.

All I want for Christmas is the best Variant Caller.

The best varian caller is YOU!

You too can be a **Variant Caller**!

We use software baseed variant callers because we can't check everything ourselves.

There are just too many variants to look at.

If you know what you are looking for, you can do a better job with pileups and by eye.

A variant caller weighs data by statistical measures that cannot incorporate your knowledge and expectations.

What are phased variants?

When there is more than one copy knowing which variants are on the same DNA is called "phasing."

Two variants are in phase if they form the same haplotype (inherited together).

"Inherited together" is misleading since they aren't always inherited together (chromosomal recombination can break that). Most often though they do.

Unphased variants are genotypes where we don't know which chromosomes hold that allele.

"Naive" Variant calling

Find short variants directly from the pileup

Works very well if the alignments are "correct" and the genome is "densely packed" with information - few redundant segments.

Bacteria, viruses in general (but not always).

It becomes an issue of statistics - how many variants would you need to see to trust it.

It is also a matter of population: clonal or not. How many alleles could there be?

If you only need to check a few variants

You can visually look them up and figure out reasonably well yourself what they are.

Especially true if your problem is subtle, and has unexpected qualities to it.

Your eye will win out over any variant caller - if you are even a little bit trained in what to do.

Typical workflow

1. Use a software to locate potential variants
2. Verify by eye the variants, especially those of importance

What if alignments are "incorrect"?

Important: alignments are **correct mathematically**.

Reality may not not conform to the scoring matrix that you chose.

You then may need to **inspect**, **correct** or **recalibrate** the alignment or results.

Why did we spend so much time with filtering?

Just about every analysis will depend on the alignment result.

In simple cases using the original BAM file may suffice.

Often it is essential that your input BAM file contains only the data that is appropriate for the analysis.

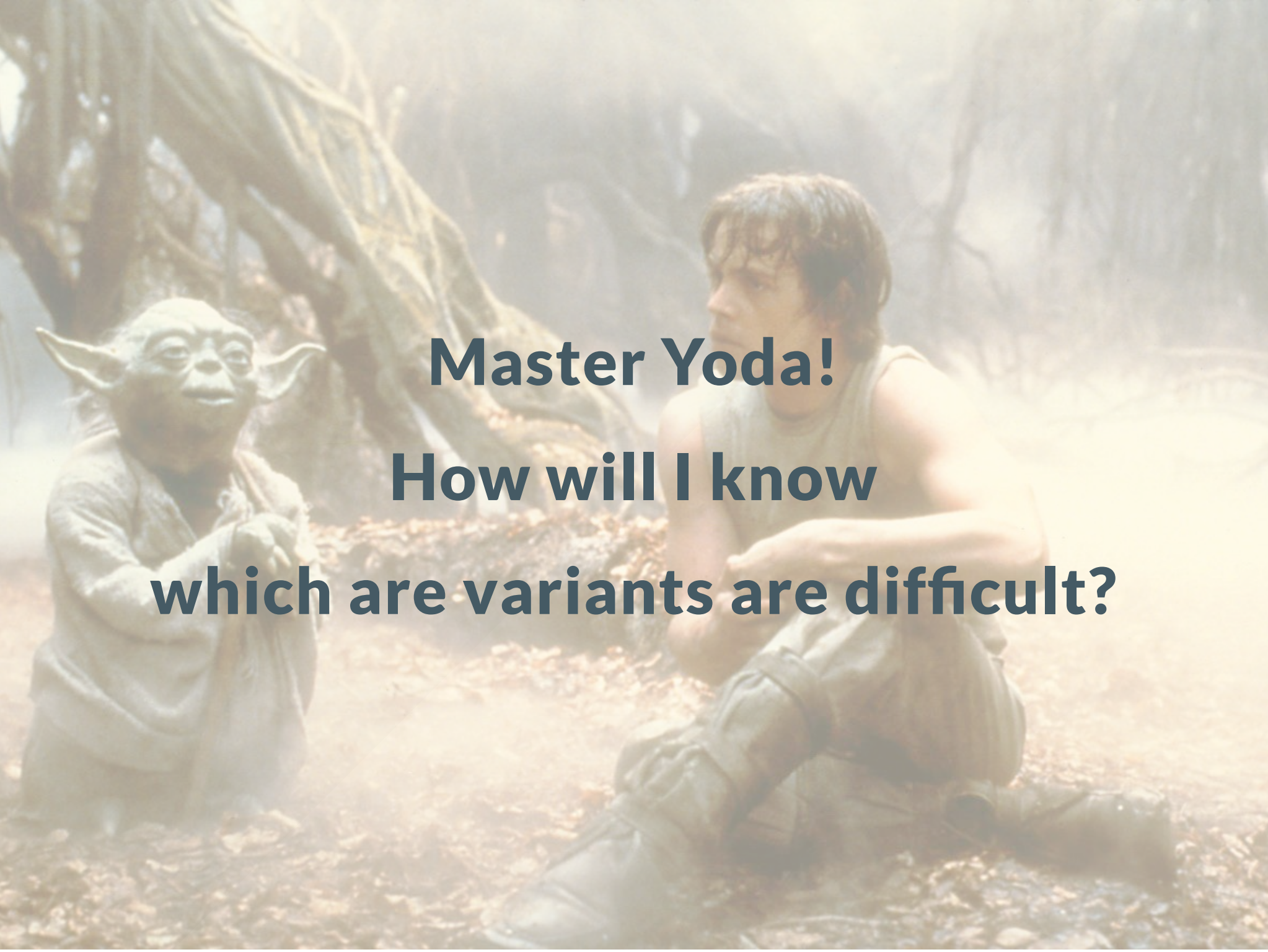
You must not be "afraid" of the BAM file.

What is the right way to call SNPs?

The answer depends on the complexity of the information in the genome.

There are several tacit and built-in assumptions that each software developer builds into their tool.

- Easy regions: can be solved by any approach
- Not so easy regions: some results are incorrect
- Difficult regions: need manual supervision


Master Yoda!
How will I know
which are variants are difficult?


Hard to call
difficult variants are!

Expectations

For example for human genome:

Doing an 80% job (80% of variants are correct and do in fact exist) is **"easy"**.

Easy here means there are documented parameter settings that do a good job.

Doing a 100% accuracte job is impossible.

Your results will typically fall somewhere between the two: 80%-100%

Let's call SNPs: Generate a mutated genome

Simulate reads with known mutations:

```
REF=refs/AF086833.fa
wgsim -N 2500 $REF read1.fq read2.fq > mutations.txt
```

The mutations file has (the simulation is diploid):

```
cat mutations.txt
AF086833.2 46 - G -
AF086833.2 174 GG - -
...
AF086833.2 2791 - GG -
...
```

Can we find these changes?

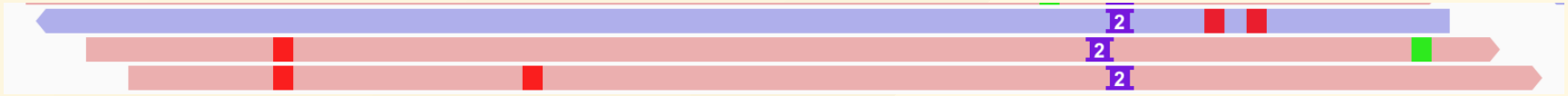
Let's call SNPs: Generate the alignments

Align the reads:

```
BAM=alignment.bam
bwa mem $REF read1.fq read2.fq | samtools sort > $BAM
samtools index $BAM
```

What is a misalignment?

Remember that your simulation might be different. We zoom in closer over two insertions



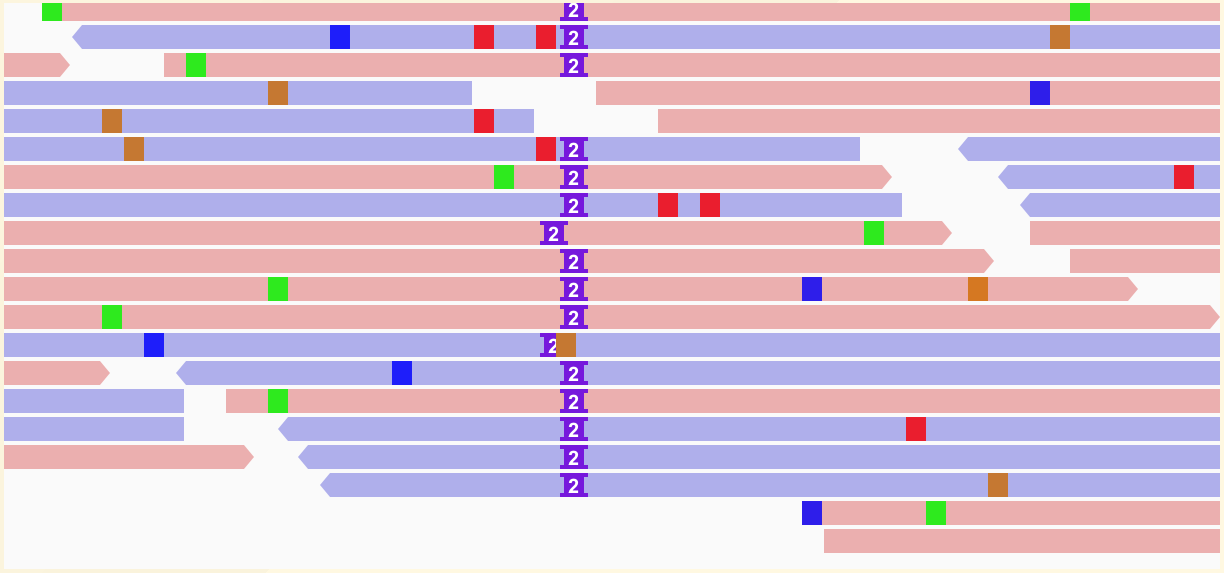
See how the middle read indicates the insertion in the "wrong" position?

Also, note how otherwise it is **perfectly matching** around the INDEL!

It is "misaligned" but not through the fault of the aligner. We could **fix** the middle read by looking at the other reads around it: **realignment**

The Ebola genome is information dense

In this case, there are plenty of "correct" alignments that will "rescue" the incorrect one.



This is not always the case. You could have the majority or reads misalign --> **incorrect calls**

What is complex and what is simple?

In information technology:

- **Complex** usually means lots of information.
- **Simple*** usually means little information.

In bioinformatics, it is easier to solve genomes with lots of information than genomes with little information.

Low information content means redundant elements, repetitive regions, low complexity base content.

Bioinformaticians inverted the meaning of the word "complex."

What we call "complex" means difficult to solve because the information is thin, spread out across lots of bases.

So the Ebola genome is "simple."

That means it is dense in information, and that makes it less sensitive to scoring matrices.

The more "complex" a genome, the more likely is that the alignments go awry.

Think about it as not having enough information to **anchor** a read to the right location.

Accurately locating the origin becomes even more challenging when the read does not even match the reference because of the mutation.

Let's call SNPs: how to I pick the right tool?

Different people swear by various tools.

Some tools are tuned to some specific use cases.

`GATK` for human/cancer genome - but it is not a tool - more of a lifestyle - one that happens to be unecessarily complicated and obtuse. You must bend the knee when you use `GATK`.

We will demonstrate a few other approaches:

- `bcftools`
- `freebayes`

Let's call SNPs: using `bcftools`

A two step process:

```
# Compute the genotype likelihoods for each base.
bcftools mpileup -Ovu -f $REF $BAM > gt.vcf
```

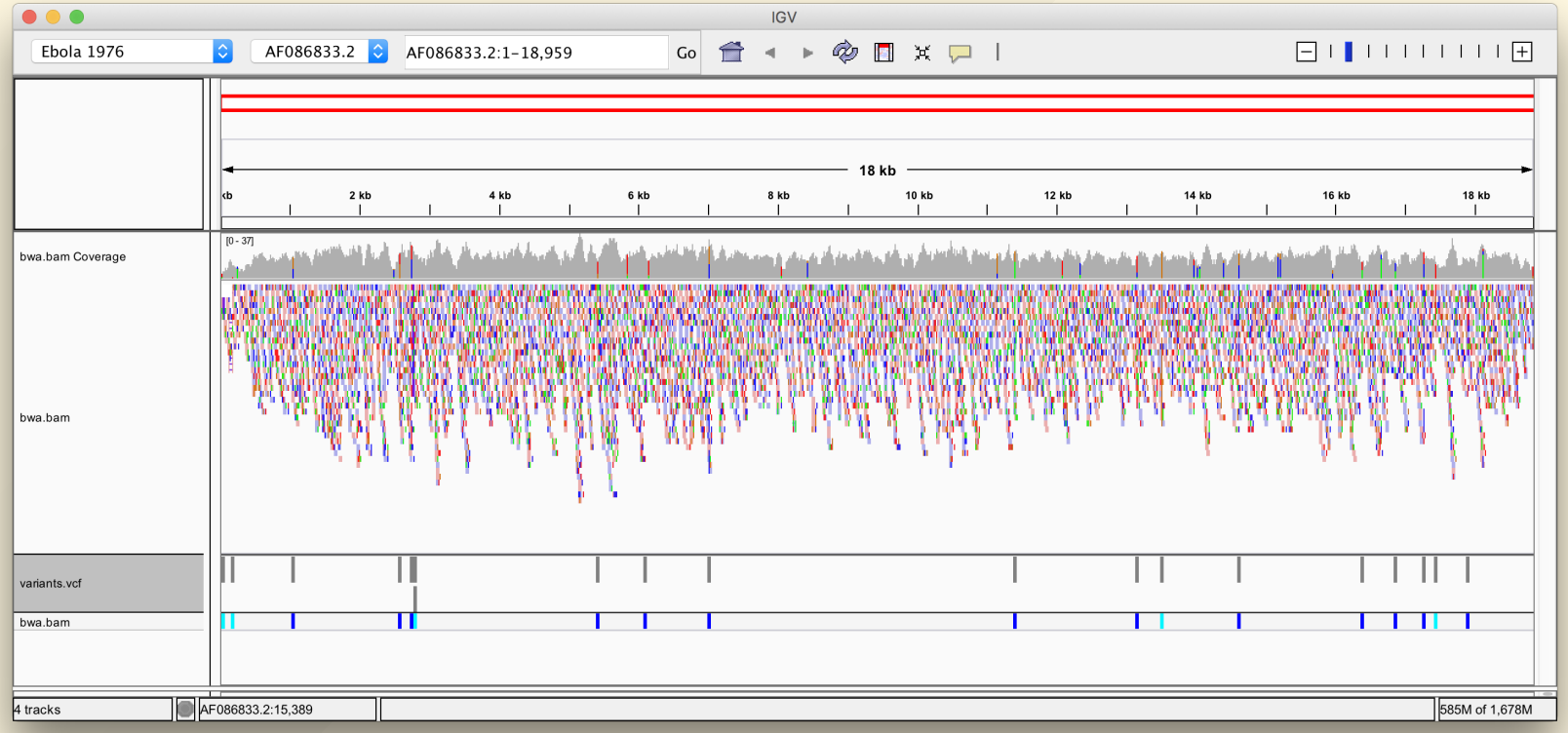
VCF files are explained in other lectures. Just take them as they are here.

```
# Call the variants. The simulation is diploid!
bcftools call -vm -Ov gt.vcf > variants1.vcf
```

The final results of this sequencing experiment are in `variants.vcf`. This is the file that needs a biological interpretation.

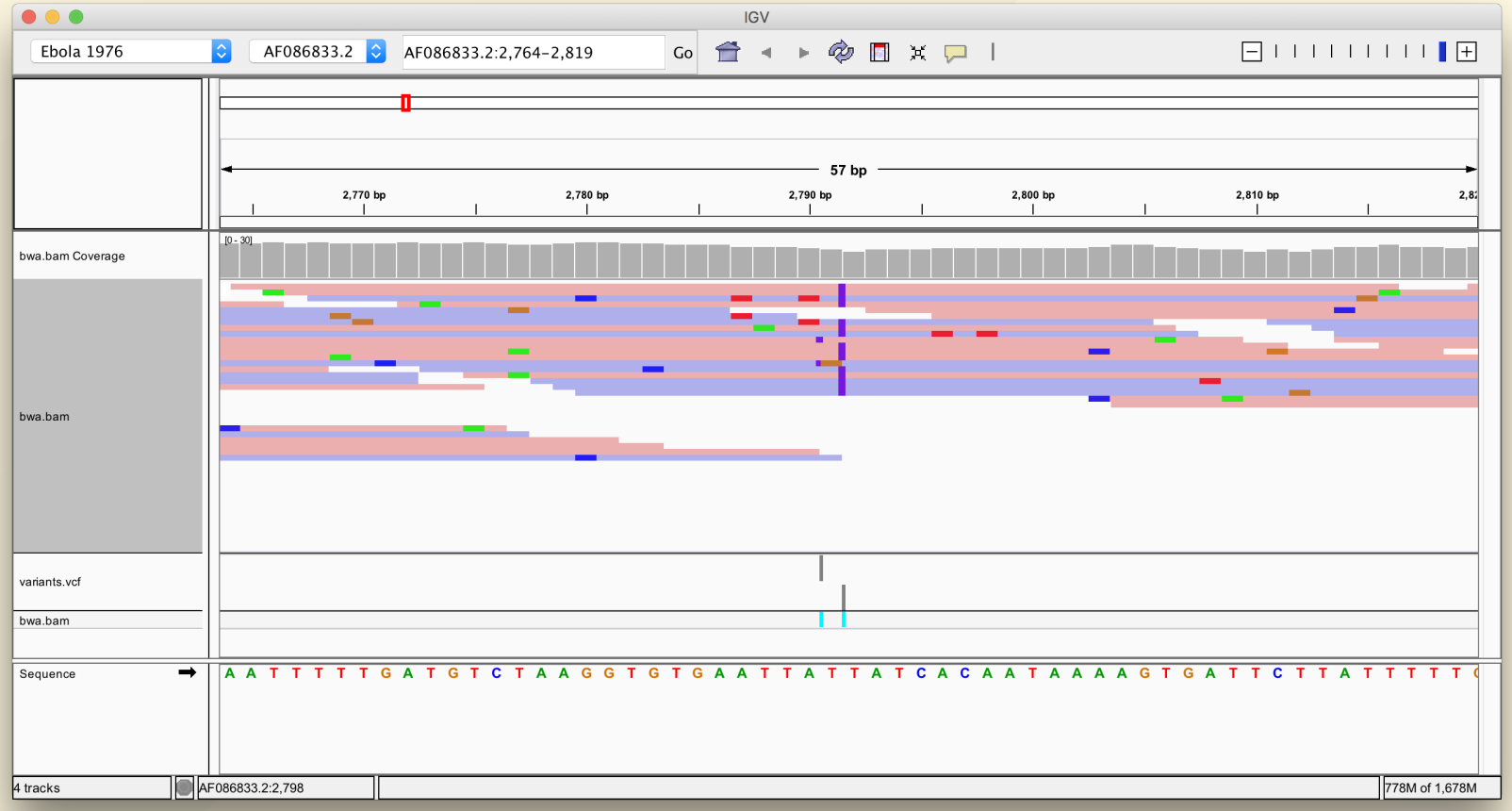
Let's call SNPs: interpreting the output

The bottom tracks are the variant calls:



Let's call SNPs: did it get the funky part right?

Well, ahem...no



Let's call SNPs: using `freebayes`

A different tool, with a better understanding of misalignments:

```
freebayes -f $REF $BAM > variants2.vcf
```

Let's call SNPs: visualizing `freebayes`

Freebayes has called the tricky variation correctly



Observations

This does not necessarily mean that `freebayes` is always better.

Tools built for a specific task usually perform better - their developers understand the pitfalls of that job and have builtin corrections.

All tools will fail in some cases.

Warning sign: variants close by one another (10-30 bases or fewer)